

Design Document

Personal Finance Manager

Console Application

Clean Code Learning Assignment

Prepared as part of structured software design evaluation

Document Version: 2.1

Date: May 11, 2026

Author: Vishwas Vijayvargiya

Project: Personal Finance Manager

Type: Console-based finance manager with API + DB backend

Contents

1	Introduction	3
1.1	Purpose	3
1.2	Objective	3
1.3	Functional Scope	3
2	System Overview	3
2.1	Architecture	3
2.2	Use Case View	3
2.3	Technology Stack	5
3	Module and Class Design	5
3.1	Module Breakdown	5
3.2	Design Patterns Used	5
3.3	UML Class Diagram	6
4	Database Design	6
4.1	Overview	6
4.2	Database Schema Diagram	7
4.3	Core Table Intent	7
4.4	Budget and Summary Readiness	8
4.5	Entity-Relationship Diagram	9
5	Application Flow	10
5.1	Overview	10
5.2	Flow Coverage by Feature	11
6	Sequence Diagram	12
6.1	Add Expense – Full Interaction Sequence	12
6.2	Sequence Walk-through	13
7	Error Handling Strategy	13
7.1	Validation Layer	13
7.2	Service and API Handling	13
7.3	Persistence Safety	13
8	Clean Code Adherence	14
9	Test Case Design	14
9.1	Mandatory Scope	14
9.2	Expense Management	14
9.3	Income Management	15
9.4	Budget Management	15
9.5	Summary, Persistence, and Failures	15
10	Setup Instructions	15
11	Evaluation Criteria Mapping	16

Constraints Compliance	16
Appendix: Diagram Index	16

1 Introduction

1.1 Purpose

This document describes the software design of the **Personal Finance Manager**, developed for the Clean Code Learning Assignment. It captures architecture, module structure, database design, application flow, and testing strategy for all required features: Expense, Income, Budget, and Summary.

1.2 Objective

Build a finance management solution that demonstrates:

- clean coding practices,
- structured problem-solving,
- scalable and modular design,
- database-backed persistence across restarts.

1.3 Functional Scope

1. **Expense Management**: add, view, filter (category/date), delete.
2. **Income Management**: add and view income.
3. **Budget Management**: set and track budgets.
4. **Summary**: total income, total expenses, balance, and category-wise spending.

2 System Overview

2.1 Architecture

The solution follows layered architecture with clear separation of concerns:

1. **Presentation Layer** – Console UI and menu-driven interaction.
2. **Business Layer** – Service classes and use-case orchestration.
3. **Data Access Layer** – Repository pattern on top of EF Core.
4. **Persistence Layer** – Relational DB with entity configuration and constraints.

2.2 Use Case View

The following diagram summarizes the main user-visible capabilities supported by the application.

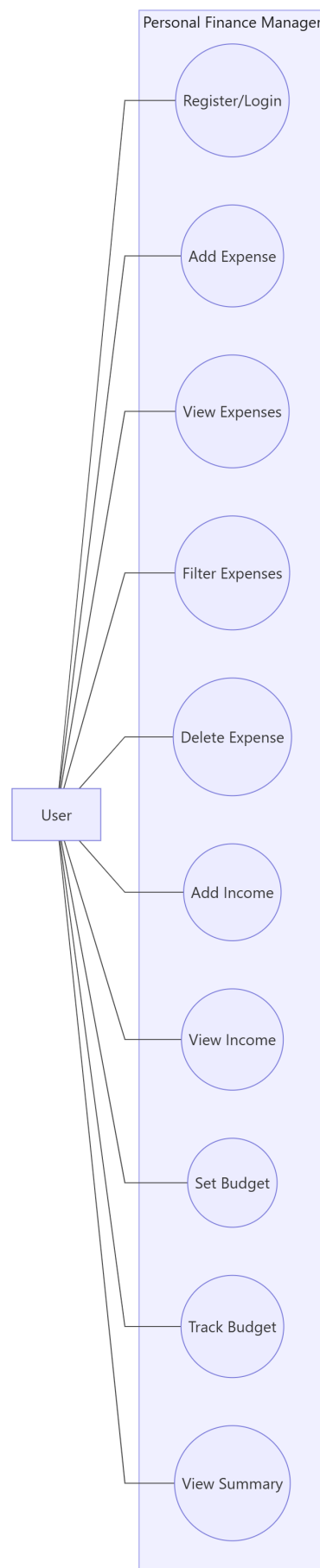


Figure 1: Use Case Diagram

2.3 Technology Stack

Component	Choice
Language / Run-time	C# / .NET
Interface	Console interaction
Backend	ASP.NET Core Web API
ORM	Entity Framework Core
Database	Relational database (SQL Server)
Authentication	JWT + refresh token flow
DI	Microsoft.Extensions.DependencyInjection

3 Module and Class Design

3.1 Module Breakdown

Module / Namespace	Responsibility
ConsoleUi / Controllers	Menu navigation and feature-specific interaction (Auth, Expense, Income, Budget, Summary)
Application / Services	Use-case orchestration and business logic contracts
Repositories	Generic CRUD + domain-specific query operations
Models / DTOs	Request/response models and domain entities
Data	AppDbContext + Fluent API configurations
Helpers / Validators	Reusable input validation, formatting, and utility helpers

3.2 Design Patterns Used

- **Repository Pattern:** decouples persistence from business logic.
- **Service Layer:** keeps controllers/UI thin and maintainable.
- **Dependency Injection:** interfaces + DI for loose coupling.
- **Generic Base Abstractions:** reusable CRUD service/repository classes.

```

classDiagram
    class Program {
        +Name()
    }
    class App {
        +StartApp()
        +QuitAppApp()
    }
    class Auth {
        +AuthenticateUser()
        +RegisterUser()
        +VerifyToken()
    }
    class Register {
        +CreateUserAndToken()
    }
    class Login {
        +AuthenticateUserAndToken()
    }
    class Logout {
        +RevokeToken()
    }
    class Profile {
        +GetUserProfile()
    }
    class Settings {
        +GetUserSettings()
        +UpdateUserSettings()
    }
    class Dashboard {
        +GetDashboardData()
    }
    class Reports {
        +GenerateReport()
    }
    class Analytics {
        +GetAnalyticsData()
    }
    class Notifications {
        +GetNotifications()
    }
    class Help {
        +GetHelpContent()
    }
    class Admin {
        +ManageUsers()
        +ManageRoles()
    }
    class Audit {
        +GetAuditLog()
    }
    class Backup {
        +PerformBackup()
    }
    class Restore {
        +PerformRestore()
    }
    class Upgrade {
        +CheckForUpdates()
    }
    class Security {
        +PerformSecurityScan()
    }
    class Compliance {
        +GenerateComplianceReport()
    }
    class Integration {
        +ConnectExternalService()
    }
    class Localization {
        +GetLocalizationStrings()
    }
    class Performance {
        +MonitorPerformance()
    }
    class Troubleshooting {
        +GetTroubleshootingSteps()
    }
    class Feedback {
        +SubmitFeedback()
    }
    class Support {
        +GetSupportTickets()
    }
    class Legal {
        +GetLegalDocuments()
    }
    class Privacy {
        +GetPrivacyPolicy()
    }
    class Terms {
        +GetTermsOfService()
    }
    class Cookies {
        +GetCookiePolicy()
    }
    class Accessibility {
        +GetAccessibilityStatement()
    }
    class Sustainability {
        +GetSustainabilityReport()
    }
    class Diversity {
        +GetDiversityStatement()
    }
    class Ethics {
        +GetEthicsStatement()
    }
    class Governance {
        +GetGovernanceFramework()
    }
    class Risk {
        +GetRiskAssessment()
    }
    class Resilience {
        +GetResiliencePlan()
    }
    class Innovation {
        +GetInnovationStrategy()
    }
    class Research {
        +GetResearchInitiatives()
    }
    class Development {
        +GetDevelopmentRoadmap()
    }
    class Testing {
        +GetTestingStrategy()
    }
    class Deployment {
        +GetDeploymentPlan()
    }
    class Monitoring {
        +GetMonitoringDashboard()
    }
    class Maintenance {
        +GetMaintenanceSchedule()
    }
    class Updates {
        +GetUpdateLog()
    }
    class Patches {
        +GetPatchNotes()
    }
    class SecurityUpdates {
        +GetSecurityUpdateInfo()
    }
    class PerformanceUpdates {
        +GetPerformanceUpdateInfo()
    }
    class BugFixes {
        +GetBugFixInfo()
    }
    class NewFeatures {
        +GetNewFeatureInfo()
    }
    class EndOfLife {
        +GetEndOfLifeInfo()
    }
    class Discontinuation {
        +GetDiscontinuationInfo()
    }
    class Migration {
        +GetMigrationPlan()
    }
    class Archival {
        +GetArchivalPolicy()
    }
    class Preservation {
        +GetPreservationPolicy()
    }
    class Accession {
        +GetAccessionPolicy()
    }
    class Digitization {
        +GetDigitizationPolicy()
    }
    class DigitizationPolicies {
        +GetDigitizationPolicies()
    }
    class DigitizationStandards {
        +GetDigitizationStandards()
    }
    class DigitizationGuidelines {
        +GetDigitizationGuidelines()
    }
    class DigitizationBestPractices {
        +GetDigitizationBestPractices()
    }
    class DigitizationTools {
        +GetDigitizationTools()
    }
    class DigitizationResources {
        +GetDigitizationResources()
    }
    class DigitizationTraining {
        +GetDigitizationTraining()
    }
    class DigitizationSupport {
        +GetDigitizationSupport()
    }
    class DigitizationPartners {
        +GetDigitizationPartners()
    }
    class DigitizationFunding {
        +GetDigitizationFunding()
    }
    class DigitizationImpact {
        +GetDigitizationImpact()
    }
    class DigitizationFuture {
        +GetDigitizationFuture()
    }
    class DigitizationVision {
        +GetDigitizationVision()
    }
    class DigitizationMission {
        +GetDigitizationMission()
    }
    class DigitizationValues {
        +GetDigitizationValues()
    }
    class DigitizationPrinciples {
        +GetDigitizationPrinciples()
    }
    class DigitizationGoals {
        +GetDigitizationGoals()
    }
    class DigitizationObjectives {
        +GetDigitizationObjectives()
    }
    class DigitizationStrategies {
        +GetDigitizationStrategies()
    }
    class DigitizationTactics {
        +GetDigitizationTactics()
    }
    class DigitizationActions {
        +GetDigitizationActions()
    }
    class DigitizationResults {
        +GetDigitizationResults()
    }
    class DigitizationMetrics {
        +GetDigitizationMetrics()
    }
    class DigitizationKPIs {
        +GetDigitizationKPIs()
    }
    class DigitizationIndicators {
        +GetDigitizationIndicators()
    }
    class DigitizationBenchmarks {
        +GetDigitizationBenchmarks()
    }
    class DigitizationStandardsCompliance {
        +GetDigitizationStandardsCompliance()
    }
    class DigitizationGuidelinesCompliance {
        +GetDigitizationGuidelinesCompliance()
    }
    class DigitizationBestPracticesCompliance {
        +GetDigitizationBestPracticesCompliance()
    }
    class DigitizationToolsCompliance {
        +GetDigitizationToolsCompliance()
    }
    class DigitizationResourcesCompliance {
        +GetDigitizationResourcesCompliance()
    }
    class DigitizationTrainingCompliance {
        +GetDigitizationTrainingCompliance()
    }
    class DigitizationSupportCompliance {
        +GetDigitizationSupportCompliance()
    }
    class DigitizationPartnersCompliance {
        +GetDigitizationPartnersCompliance()
    }
    class DigitizationFundingCompliance {
        +GetDigitizationFundingCompliance()
    }
    class DigitizationImpactCompliance {
        +GetDigitizationImpactCompliance()
    }
    class DigitizationFutureCompliance {
        +GetDigitizationFutureCompliance()
    }
    class DigitizationVisionCompliance {
        +GetDigitizationVisionCompliance()
    }
    class DigitizationMissionCompliance {
        +GetDigitizationMissionCompliance()
    }
    class DigitizationValuesCompliance {
        +GetDigitizationValuesCompliance()
    }
    class DigitizationPrinciplesCompliance {
        +GetDigitizationPrinciplesCompliance()
    }
    class DigitizationGoalsCompliance {
        +GetDigitizationGoalsCompliance()
    }
    class DigitizationObjectivesCompliance {
        +GetDigitizationObjectivesCompliance()
    }
    class DigitizationStrategiesCompliance {
        +GetDigitizationStrategiesCompliance()
    }
    class DigitizationTacticsCompliance {
        +GetDigitizationTacticsCompliance()
    }
    class DigitizationActionsCompliance {
        +GetDigitizationActionsCompliance()
    }
    class DigitizationResultsCompliance {
        +GetDigitizationResultsCompliance()
    }
    class DigitizationMetricsCompliance {
        +GetDigitizationMetricsCompliance()
    }
    class DigitizationKPIsCompliance {
        +GetDigitizationKPIsCompliance()
    }
    class DigitizationIndicatorsCompliance {
        +GetDigitizationIndicatorsCompliance()
    }
    class DigitizationBenchmarksCompliance {
        +GetDigitizationBenchmarksCompliance()
    }
    class DigitizationStandardsComplianceReport {
        +GetDigitizationStandardsComplianceReport()
    }
    class DigitizationGuidelinesComplianceReport {
        +GetDigitizationGuidelinesComplianceReport()
    }
    class DigitizationBestPracticesComplianceReport {
        +GetDigitizationBestPracticesComplianceReport()
    }
    class DigitizationToolsComplianceReport {
        +GetDigitizationToolsComplianceReport()
    }
    class DigitizationResourcesComplianceReport {
        +GetDigitizationResourcesComplianceReport()
    }
    class DigitizationTrainingComplianceReport {
        +GetDigitizationTrainingComplianceReport()
    }
    class DigitizationSupportComplianceReport {
        +GetDigitizationSupportComplianceReport()
    }
    class DigitizationPartnersComplianceReport {
        +GetDigitizationPartnersComplianceReport()
    }
    class DigitizationFundingComplianceReport {
        +GetDigitizationFundingComplianceReport()
    }
    class DigitizationImpactComplianceReport {
        +GetDigitizationImpactComplianceReport()
    }
    class DigitizationFutureComplianceReport {
        +GetDigitizationFutureComplianceReport()
    }
    class DigitizationVisionComplianceReport {
        +GetDigitizationVisionComplianceReport()
    }
    class DigitizationMissionComplianceReport {
        +GetDigitizationMissionComplianceReport()
    }
    class DigitizationValuesComplianceReport {
        +GetDigitizationValuesComplianceReport()
    }
    class DigitizationPrinciplesComplianceReport {
        +GetDigitizationPrinciplesComplianceReport()
    }
    class DigitizationGoalsComplianceReport {
        +GetDigitizationGoalsComplianceReport()
    }
    class DigitizationObjectivesComplianceReport {
        +GetDigitizationObjectivesComplianceReport()
    }
    class DigitizationStrategiesComplianceReport {
        +GetDigitizationStrategiesComplianceReport()
    }
    class DigitizationTacticsComplianceReport {
        +GetDigitizationTacticsComplianceReport()
    }
    class DigitizationActionsComplianceReport {
        +GetDigitizationActionsComplianceReport()
    }
    class DigitizationResultsComplianceReport {
        +GetDigitizationResultsComplianceReport()
    }
    class DigitizationMetricsComplianceReport {
        +GetDigitizationMetricsComplianceReport()
    }
    class DigitizationKPIsComplianceReport {
        +GetDigitizationKPIsComplianceReport()
    }
    class DigitizationIndicatorsComplianceReport {
        +GetDigitizationIndicatorsComplianceReport()
    }
    class DigitizationBenchmarksComplianceReport {
        +GetDigitizationBenchmarksComplianceReport()
    }
    class DigitizationStandardsComplianceReportGenerator {
        +GenerateDigitizationStandardsComplianceReport()
    }
    class DigitizationGuidelinesComplianceReportGenerator {
        +GenerateDigitizationGuidelinesComplianceReport()
    }
    class DigitizationBestPracticesComplianceReportGenerator {
        +GenerateDigitizationBestPracticesComplianceReport()
    }
    class DigitizationToolsComplianceReportGenerator {
        +GenerateDigitizationToolsComplianceReport()
    }
    class DigitizationResourcesComplianceReportGenerator {
        +GenerateDigitizationResourcesComplianceReport()
    }
    class DigitizationTrainingComplianceReportGenerator {
        +GenerateDigitizationTrainingComplianceReport()
    }
    class DigitizationSupportComplianceReportGenerator {
        +GenerateDigitizationSupportComplianceReport()
    }
    class DigitizationPartnersComplianceReportGenerator {
        +GenerateDigitizationPartnersComplianceReport()
    }
    class DigitizationFundingComplianceReportGenerator {
        +GenerateDigitizationFundingComplianceReport()
    }
    class DigitizationImpactComplianceReportGenerator {
        +GenerateDigitizationImpactComplianceReport()
    }
    class DigitizationFutureComplianceReportGenerator {
        +GenerateDigitizationFutureComplianceReport()
    }
    class DigitizationVisionComplianceReportGenerator {
        +GenerateDigitizationVisionComplianceReport()
    }
    class DigitizationMissionComplianceReportGenerator {
        +GenerateDigitizationMissionComplianceReport()
    }
    class DigitizationValuesComplianceReportGenerator {
        +GenerateDigitizationValuesComplianceReport()
    }
    class DigitizationPrinciplesComplianceReportGenerator {
        +GenerateDigitizationPrinciplesComplianceReport()
    }
    class DigitizationGoalsComplianceReportGenerator {
        +GenerateDigitizationGoalsComplianceReport()
    }
    class DigitizationObjectivesComplianceReportGenerator {
        +GenerateDigitizationObjectivesComplianceReport()
    }
    class DigitizationStrategiesComplianceReportGenerator {
        +GenerateDigitizationStrategiesComplianceReport()
    }
    class DigitizationTacticsComplianceReportGenerator {
        +GenerateDigitizationTacticsComplianceReport()
    }
    class DigitizationActionsComplianceReportGenerator {
        +GenerateDigitizationActionsComplianceReport()
    }
    class DigitizationResultsComplianceReportGenerator {
        +GenerateDigitizationResultsComplianceReport()
    }
    class DigitizationMetricsComplianceReportGenerator {
        +GenerateDigitizationMetricsComplianceReport()
    }
    class DigitizationKPIsComplianceReportGenerator {
        +GenerateDigitizationKPIsComplianceReport()
    }
    class DigitizationIndicatorsComplianceReportGenerator {
        +GenerateDigitizationIndicatorsComplianceReport()
    }
    class DigitizationBenchmarksComplianceReportGenerator {
        +GenerateDigitizationBenchmarksComplianceReport()
    }
    class DigitizationStandardsComplianceReportGenerator {
        +GenerateDigitizationStandardsComplianceReport()
    }
    class DigitizationGuidelinesComplianceReportGenerator {
        +GenerateDigitizationGuidelinesComplianceReport()
    }
    class DigitizationBestPracticesComplianceReportGenerator {
        +GenerateDigitizationBestPracticesComplianceReport()
    }
    class DigitizationToolsComplianceReportGenerator {
        +GenerateDigitizationToolsComplianceReport()
    }
    class DigitizationResourcesComplianceReportGenerator {
        +GenerateDigitizationResourcesComplianceReport()
    }
    class DigitizationTrainingComplianceReportGenerator {
        +GenerateDigitizationTrainingComplianceReport()
    }
    class DigitizationSupportComplianceReportGenerator {
        +GenerateDigitizationSupportComplianceReport()
    }
    class DigitizationPartnersComplianceReportGenerator {
        +GenerateDigitizationPartnersComplianceReport()
    }
    class DigitizationFundingComplianceReportGenerator {
        +GenerateDigitizationFundingComplianceReport()
    }
    class DigitizationImpactComplianceReportGenerator {
        +GenerateDigitizationImpactComplianceReport()
    }
    class DigitizationFutureComplianceReportGenerator {
        +GenerateDigitizationFutureComplianceReport()
    }
    class DigitizationVisionComplianceReportGenerator {
        +GenerateDigitizationVisionComplianceReport()
    }
    class DigitizationMissionComplianceReportGenerator {
        +GenerateDigitizationMissionComplianceReport()
    }
    class DigitizationValuesComplianceReportGenerator {
        +GenerateDigitizationValuesComplianceReport()
    }
    class DigitizationPrinciplesComplianceReportGenerator {
        +GenerateDigitizationPrinciplesComplianceReport()
    }
    class DigitizationGoalsComplianceReportGenerator {
        +GenerateDigitizationGoalsComplianceReport()
    }
    class DigitizationObjectivesComplianceReportGenerator {
        +GenerateDigitizationObjectivesComplianceReport()
    }
    class DigitizationStrategiesComplianceReportGenerator {
        +GenerateDigitizationStrategiesComplianceReport()
    }
    class DigitizationTacticsComplianceReportGenerator {
        +GenerateDigitizationTacticsComplianceReport()
    }
    class DigitizationActionsComplianceReportGenerator {
        +GenerateDigitizationActionsComplianceReport()
    }
    class DigitizationResultsComplianceReportGenerator {
        +GenerateDigitizationResultsComplianceReport()
    }
    class DigitizationMetricsComplianceReportGenerator {
        +GenerateDigitizationMetricsComplianceReport()
    }
    class DigitizationKPIsComplianceReportGenerator {
        +GenerateDigitizationKPIsComplianceReport()
    }
    class DigitizationIndicatorsComplianceReportGenerator {
        +GenerateDigitizationIndicatorsComplianceReport()
    }
    class DigitizationBenchmarksComplianceReportGenerator {
        +GenerateDigitizationBenchmarksComplianceReport()
    }
    class DigitizationStandardsComplianceReportGenerator {
        +GenerateDigitizationStandardsComplianceReport()
    }
    class DigitizationGuidelinesComplianceReportGenerator {
        +GenerateDigitizationGuidelinesComplianceReport()
    }
    class DigitizationBestPracticesComplianceReportGenerator {
        +GenerateDigitizationBestPracticesComplianceReport()
    }
    class DigitizationToolsComplianceReportGenerator {
        +GenerateDigitizationToolsComplianceReport()
    }
    class DigitizationResources
```

```

classDiagram
    class Propagator {
        +id()
    }
    class AspController {
        +get()
        +get()
        +get()
    }
    class DependentController {
        +getExpenseById()
        +getExpenseById()
        +getExpenseById()
    }
    class IncomeController {
        +getIncomeById()
        +getIncomeById()
        +getIncomeById()
    }
    class BudgetController {
        +getBudgetById()
        +getBudgetById()
    }
    class AspService {
    }
    class DependentService {
    }
    class IncomeService {
    }
    class BudgetService {
    }
    class DependentRepository {
    }
    class IncomeRepository {
    }
    class BudgetRepository {
    }
    class Expense {
        +id()
        +name()
        +amount()
        +category()
    }
    class Income {
        +id()
        +name()
        +amount()
        +category()
    }
    class Budget {
        +id()
        +name()
        +amount()
        +category()
    }

    Propagator --> AspController
    Propagator --> DependentController
    Propagator --> IncomeController
    Propagator --> BudgetController
    AspController --> AspService
    DependentController --> DependentService
    IncomeController --> IncomeService
    BudgetController --> BudgetService
    DependentService --> DependentRepository
    IncomeService --> IncomeRepository
    BudgetService --> BudgetRepository
    DependentRepository --> Expense
    IncomeRepository --> Income
    BudgetRepository --> Budget
    
```

Figure 2: UML Class Diagrams

4.2 Database Schema Diagram

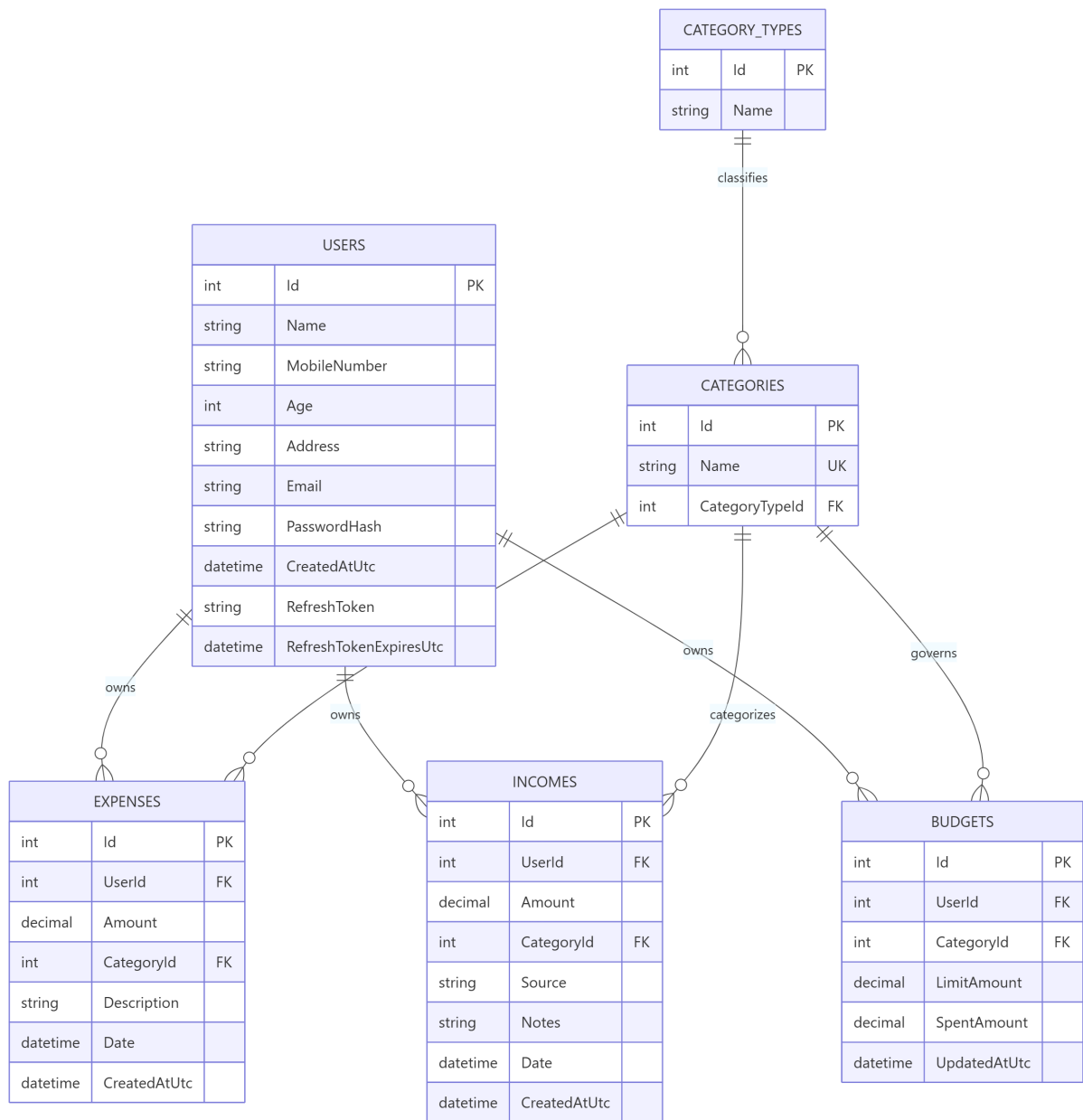


Figure 3: Relational Database Schema Diagram

4.3 Core Table Intent

- **Users:** identity and auth metadata.
- **Categories / CategoryTypes:** spending and income classification.
- **Expenses / Incomes:** monetary transactions linked to user and category.
- **Budgets:** per-user per-category limit and spent tracking.

4.4 Budget and Summary Readiness

Budget entity and constraints are part of schema so summary reporting and budget tracking can be fully supported (totals, balances, and category-wise analysis).

4.5 Entity-Relationship Diagram

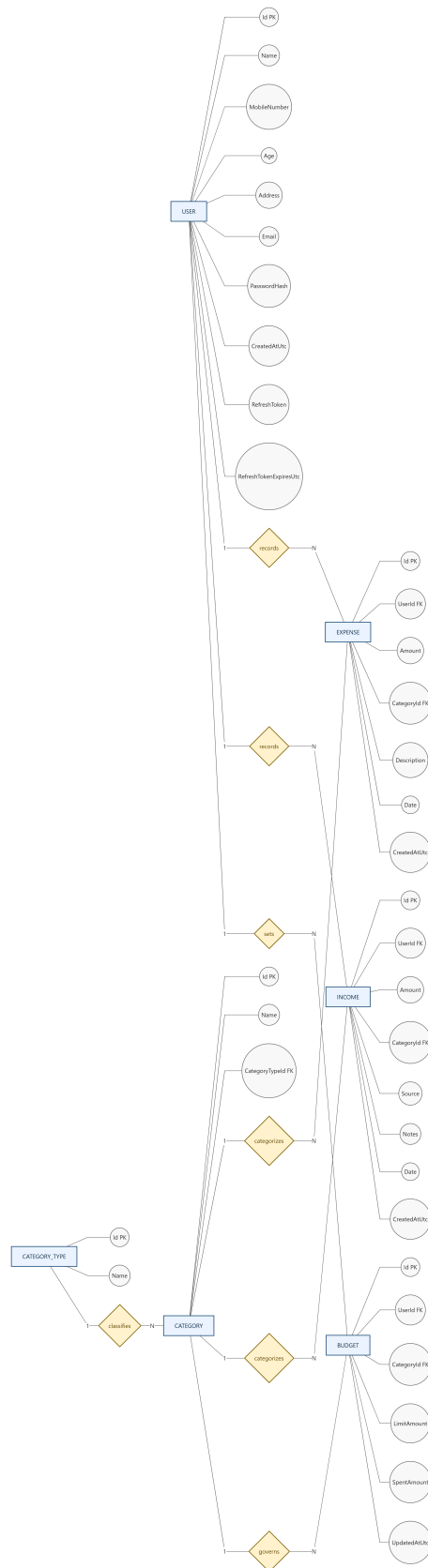
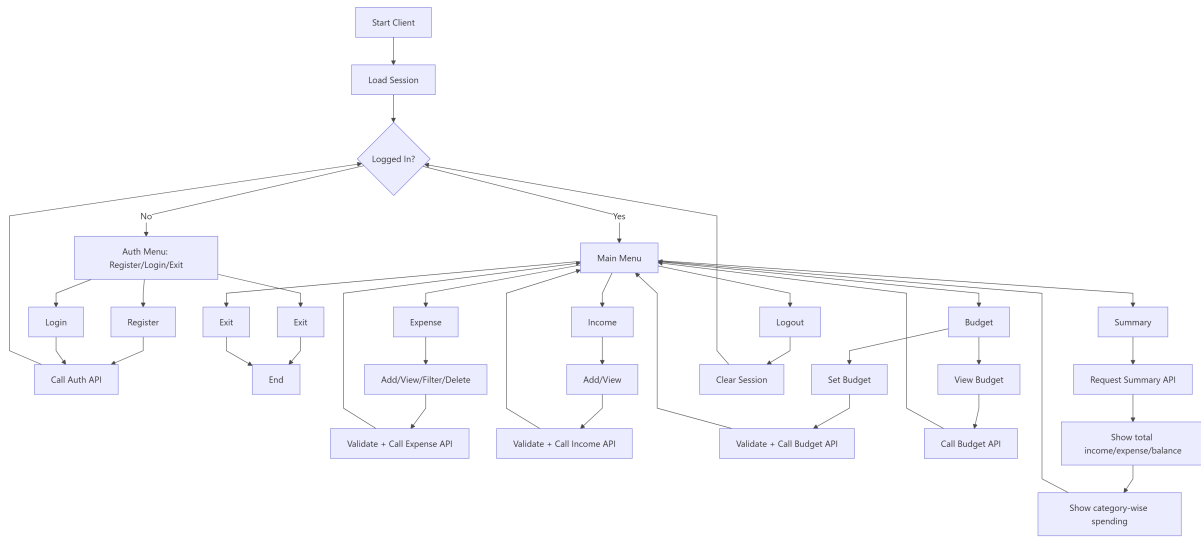


Figure 4: Entity-Relationship (ER) Diagram

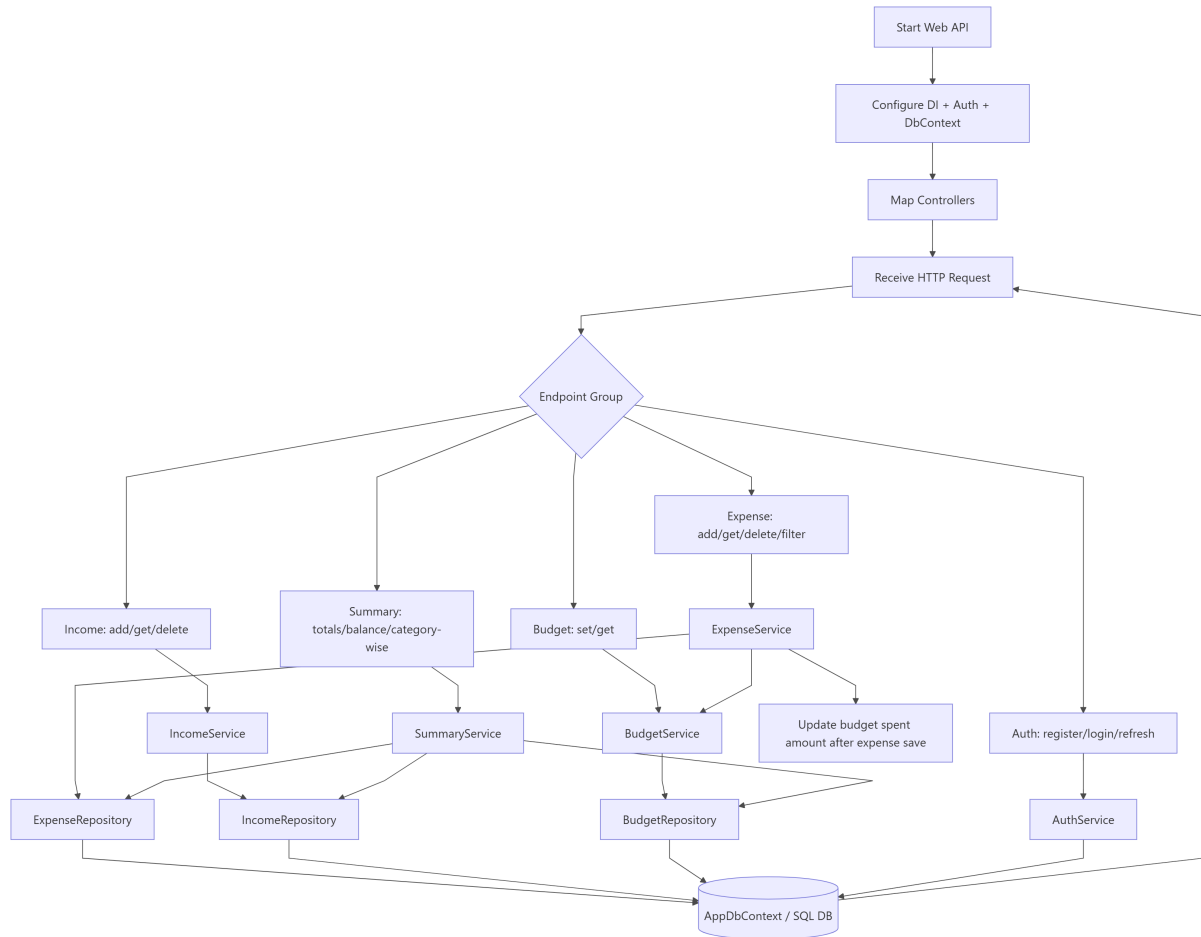
5 Application Flow

5.1 Overview

The system starts from console menu flow, authenticates user, and routes feature operations through service and repository layers to the database.



(a) Client flow



(b) Server flow

Figure 5: Application Flow Diagrams

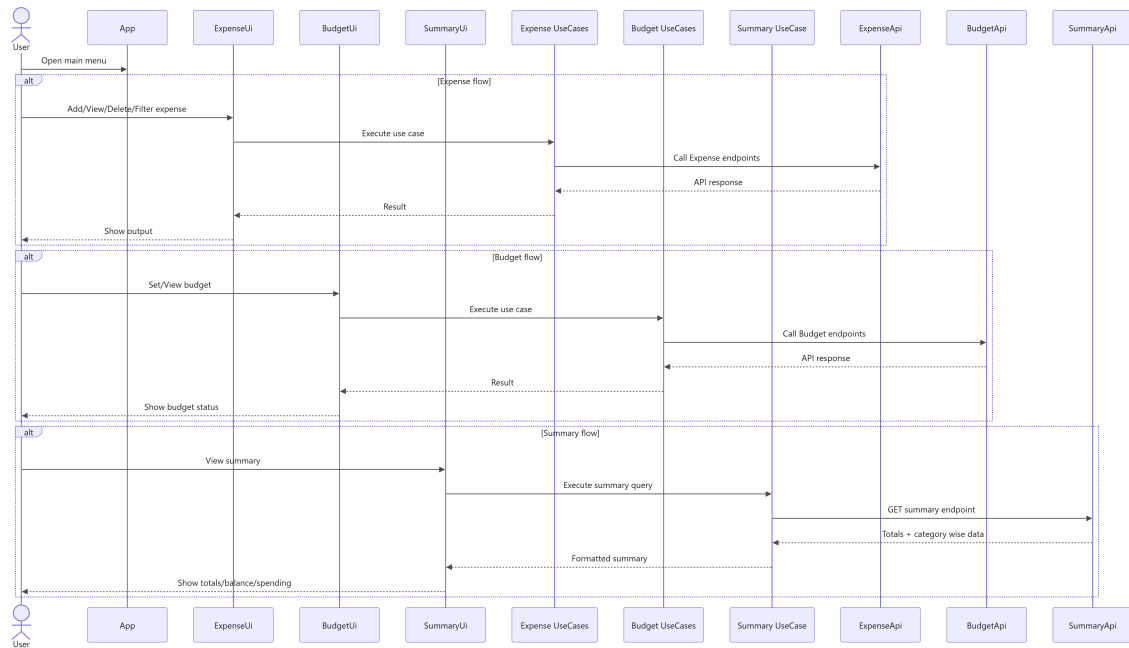
5.2 Flow Coverage by Feature

- **Expense:** add, view, filter (category/date), delete.
- **Income:** add, view.

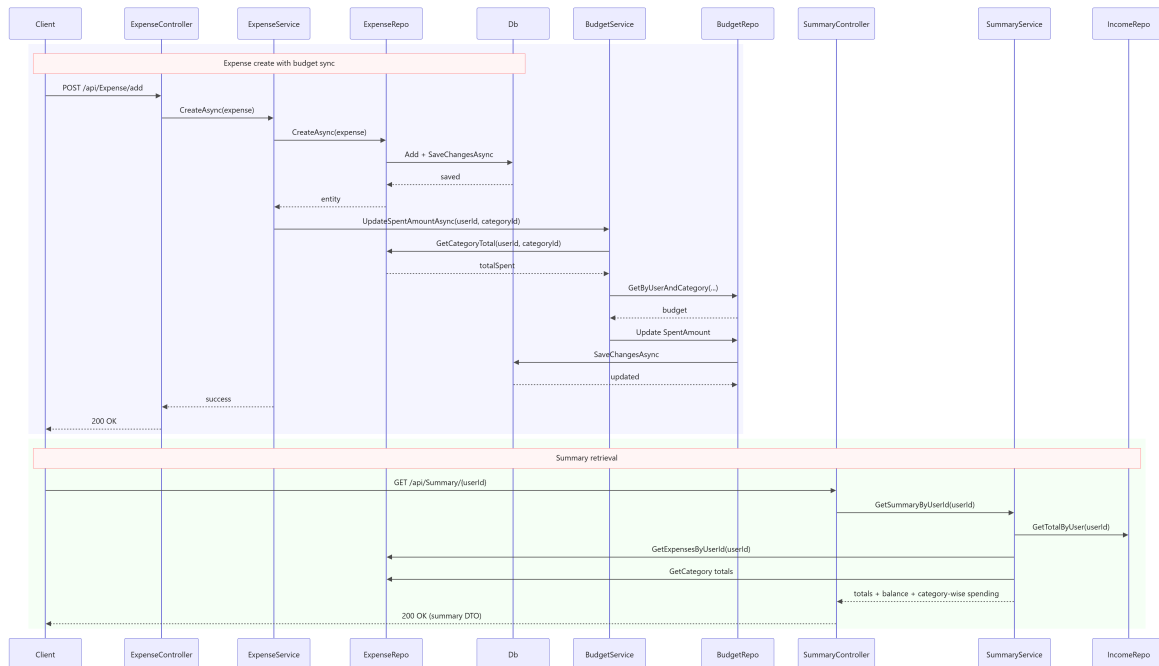
- **Budget:** set/update category budgets and track spent amount.
- **Summary:** aggregate total income, expense, balance, and category-wise spend.

6 Sequence Diagram

6.1 Add Expense – Full Interaction Sequence



(a) Client interaction sequence



(b) Server interaction sequence

Figure 6: Sequence Diagrams – Add Expense Flow

6.2 Sequence Walk-through

1. User selects Add Expense from console menu.
2. Input is validated for amount/category/date/description.
3. Service layer receives validated DTO/model.
4. Repository persists expense via EF Core.
5. Data is committed with async save.
6. Success/failure message is shown in console.
7. Budget tracking hook/flow remains part of final target behavior.

7 Error Handling Strategy

7.1 Validation Layer

- Numeric/date parsing with guarded prompts.
- Required vs optional field handling.
- Safe menu input bounds and retry prompts.

7.2 Service and API Handling

- Exceptions are converted to user-friendly output.
- Unauthorized/invalid operations return clear messages.
- Async DB and HTTP operations are wrapped with robust handling.

7.3 Persistence Safety

- Entity constraints enforce data integrity.
- Foreign key relationships maintain consistency.
- Relational persistence ensures restart-safe data.

8 Clean Code Adherence

Principle	Implementation
Meaningful Names	Descriptive classes/methods and explicit feature naming.
Small Functions	Focused methods in UI, service, and repository layers.
No Duplication	Shared CRUD logic and reusable helper abstractions.
Separation of Concerns	Presentation, business, and data concerns isolated.
Loose Coupling	Interface-based contracts and DI composition.
Maintainability	Modular boundaries support easy extension (e.g., budget/summary completion).
Error Handling	Input validation + graceful error messages and defensive checks.

9 Test Case Design

9.1 Mandatory Scope

Test scenarios cover:

- Expense Management (valid and invalid),
- Income Management (valid and invalid),
- Budget Management (valid and invalid),
- Filtering and query scenarios,
- Summary calculations,
- Database persistence and failure handling,
- Console input validation.

9.2 Expense Management

#	Scenario	Input	Expected Output
1	Add valid expense	Amount/category/date	Record saved valid
2	Add invalid amount	0 or negative value	Validation error
3	Filter by category/date	Valid filters	Correct subset listed
4	Delete existing expense	Valid ID	Expense removed
5	Delete invalid expense	Invalid ID	Friendly error

9.3 Income Management

#	Scenario	Input	Expected Output
6	Add valid income	Amount/source/date valid	Record saved
7	Add invalid income	Missing source/invalid amount	Validation error
8	View income list	Multiple records	Rows displayed

9.4 Budget Management

#	Scenario	Input	Expected Output
9	Set budget	Category + limit	Budget created/updated
10	Invalid budget	Limit ≤ 0	Validation/constraint error
11	Budget tracking	Expense exceeds limit	Warning/over-limit indication

9.5 Summary, Persistence, and Failures

#	Scenario	Input	Expected Output
12	Summary totals	Expense + income records	Correct totals/balance
13	Category-wise spend	Multi-category expenses	Category aggregation shown
14	Persistence after restart	Add records, restart app	Data remains
15	DB unavailability	Simulated DB failure	Graceful error handling

10 Setup Instructions

1. **Prerequisites:** Install .NET SDK and SQL Server.

2. **Clone repository:**

```
1 git clone https://github.com/<your-repo>/personal-finance-  
  manager.git  
2 cd personal-finance-manager
```

3. **Configure settings:** update connection strings and API settings in appsettings files.

4. **Apply migrations:**

```
1 dotnet ef database update
```


5. Run server:

```
1 dotnet run --project Server/src/WebApp.Api
```

6. Run client:

```
1 dotnet run --project Client/src/WebApp.Client
```

11 Evaluation Criteria Mapping

Criterion	How this design addresses it
Code Readability	Clear naming and understandable feature boundaries.
Design Quality	Layered, modular, scalable architecture.
Separation of Concerns	UI, service, and repository responsibilities are isolated.
Database Integration	Relational schema with EF Core constraints and mappings.
Maintainability	Interface-based design and reusable abstractions.
Error Handling	Validation checks and resilient failure responses.

Constraints Compliance

- No hardcoding of volatile configuration values.
- Avoidance of monolithic design via modular layering.
- Edge-case handling through validation and defensive checks.
- Coding standards and clean code practices followed.

Appendix: Diagram Index

Figure	Title	Section
1	Use Case Diagram	2.2
2	UML Class Diagrams	3.3
3	Relational Database Schema Diagram	4.2
4	Entity-Relationship Diagram	4.5
5	Application Flow Diagrams	5.1
6	Sequence Diagrams – Add Expense Flow	6.1